

Model free Reinforcement Learning::

Previously, we studied model based RL.

Algos ::

→ (i) Policy Iteration	} Dynamic Programming.
→ (ii) Value Iteration	
→ (iii) Async DP	

Q:- why model free RL?

DP is model based. we used bellman exp. eq.

Bellman Expectation Equation $\Rightarrow V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma V_{\pi}(s')]$

Our env is MDP and v_s represented by

$P(s', r | s, a)$ probability dist.

This probability dist. is however not

explicitly available usually. So we have to move towards Model free RL i.e those algos which do not use this prob. dist for calculations.

Q:- what is the idea behind model free RL?

I think the main idea is the **experience**

Experience is the sequence of State, action, reward. Agent interacts with the environment i.e takes action observes state, reward and then take other action.

So Experience is $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, \dots$

Experience can be **actual** or **simulated**.

⇒ **Actual**:- Agent interacts with the env

S.t model of env is not available.

⇒ **Simulated**: Agent interacts with the simulated env instead of actual.
It means model of env is available but prob. dist. required for DP are not available.

Tabular Methods

Remember, DP and these algs are also called tabular methods. We have small no. of states and actions. We can create state values for each state and state action values for each state action pair in tabular form.

Now let's focus on our first model free RL algo.

Monte Carlo Methods

Do you remember

Law of Large Numbers?

It says if we repeat a random exp multiple times, the average converges to expectation.

Return

We know that

$$V(s) = E[G_t | S_t = s]$$

↑
state at time t

So we can get value of a state if we find expectation of return in that state

Or

according to law of large numbers, we can also find this value if we average the returns for this state

That's the foundation of MC methods.
monte Carlo

"Average the returns for a particular state to get state value".

We can also extend RL to state action values.

"Average the returns for a

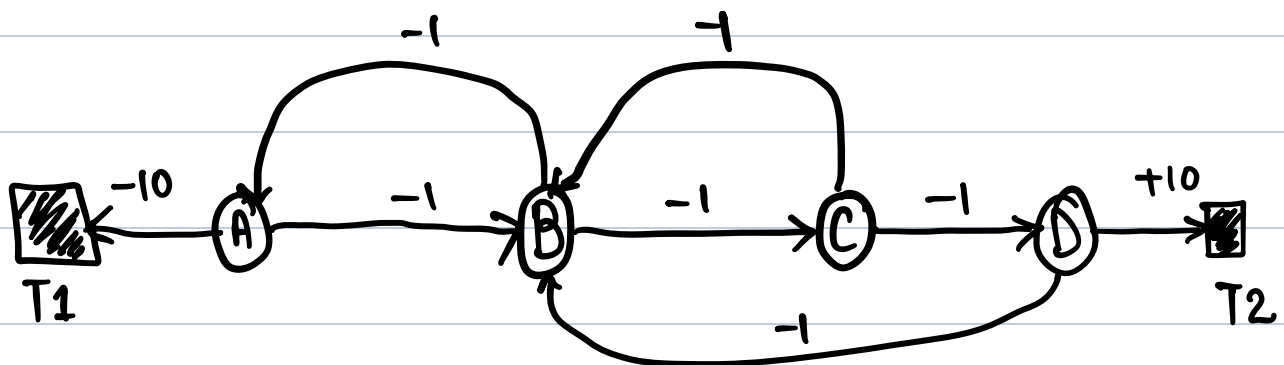
Average the returns for a particular state action pair to get state action value?

I hope you remember $V(s)$ is state value and $Q(s,a)$ is state action value.

Monte Carlo methods can be

→ First Visit Monte Carlo

→ Every Visit Monte Carlo.



For this example Env., let say we have following three episodes.

Ep-1:- A, R, -1, B, L, -1, A, L, -10.

Ep-2:- A, R, -1, B, R, -1, C, R, -1, D, R, +10.

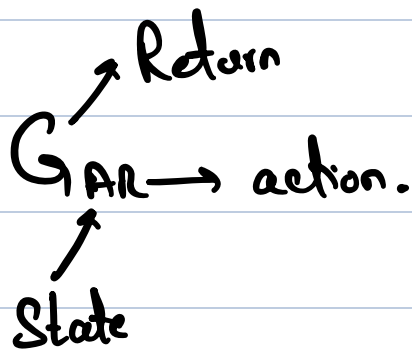
Ep-3:- B, R, -1, C, R, -1, D, L, -1, B, R, -1, C, R, -1, D, R, +10.

For each episode, returns G_t are

Ep-1:- $G_{AR} = -12$, $G_{BL} = -11$, $G_{AL} = -10$

Ep-2:- $G_{AR} = +7$, $G_{BR} = +8$, $G_{CR} = +9$, $G_{DR} = +10$

Ep-3:- $G_{BR} = +5$, $G_{CR} = +6$, $G_{DL} = +7$, $G_{BR} = +8$,
 $G_{CR} = +9$, $G_{DR} = +10$.



Lets apply first visit monte Carlo.

to find $v(A)$, average all returns for state A when it is first encountered in a trajectory/episode.

Ep-1 $G_{AR} = -12$

Ep-2 $G_{AR} = +7$

Ep-3 A not visited.

$$\text{So } V(A) = \frac{-12 + 7}{2} = -5/2 = -2.5.$$

b find $V(B)$

$$\text{Ep-1 } G_{BL} = -11$$

$$\text{Ep-2 } G_{BR} = +8$$

$$\text{Ep-3 } G_{BR} = +5$$

$$\text{So } V(B) = \frac{-11 + 8 + 5}{3} = 2/3.$$

* Points to be noted:.

→ we did not use G_{AL} from episode-1 for $V(A)$. Because, we only considered first visit

Lets find

$$Q(B, R) = ?$$

$$\text{1st Episode } G_{BR} = x$$

$$\text{2nd Episode } G_{BR} = +8$$

$$\text{3rd Episode } G_{BR} = +5$$

$$Q(B, R) = \frac{8 + 5}{2} = 6.5.$$

So in first visit monte carlo, we consider only from first encounter.

Lets try every visit mc.

$$V(A) = ?$$

Episode - I: $G_A = -12$, $G_A = -10$.

Episode - II: $G_A = +7$

Episode - III: $\times$

$$V(A) = \frac{-12 - 10 + 7}{3} = \frac{-22 + 7}{3} = \frac{-15}{3} = -5$$

Similarly, we can find Q values for state action values.

"In every visit mc, we consider return for each encounter."

So, now we know how first visit or every visit mc can be used to calculate $V(s)$ or $Q(s, a)$.

ISSUE # 1

There is a subtle point here if you noted.

"We did not encounter C, L
state action pairs in any
episode"

So we learnt nothing about its
value.

In complex systems, there may be many such
state, action pairs which are not encountered.

We know that even in DP, in control/improvement
problem we select action with max Q value. So
such actions like C, L which are not
encountered will have $Q=0$ and will be
neglected. Now, we will have to move
towards exploration.

We find value^{in evaluation} and then we select
action with maximum Q value. We act

greedy. But, now we will explore to visit (State, action) that are never encountered.

We have two approaches in this regard

* Exploring start.

probability of starting from each (S, A) is > 0 .
It means for each (S, A) pair, we will have at least one episode that starts with this pair.

But this approach is not realistic. If an env has total n states, it is not necessary that it can start from each state. For example, a car can have three states cool, heated, overheated.

It is unlikely that a car can start an episode in overheated state.

The other approach is

* ϵ -soft policy

→ when we have found $Q(s, A)$ using first or every visit MC,

→ Instead of selecting action for a particular state with max Q i.e. instead of being greedy.

we can select any action with probability ϵ/n

Exploration ↗

(n is total no. of actions).

AND

Exploitation $\Rightarrow$ we select action with max Q with probability

$$1 - \epsilon + \epsilon/n.$$

Q:- Can you prove that probabilities for exploration & exploitation sum to 1?

$$\underbrace{1 - \epsilon + \frac{\epsilon}{n}}_{\text{Exploitation}} + \underbrace{\frac{\epsilon}{n} (n-1)}_{\text{Exploration.}}$$

$$1 - \cancel{\epsilon} + \cancel{\frac{\epsilon}{n}} + \cancel{\epsilon} - \cancel{\frac{\epsilon}{n}}$$

□

→ Lets see the pseudocode

On-policy first-visit MC control (for ϵ -soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\epsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ϵ -soft policy

$Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π : $S_0, A_0, R_1, \dots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \dots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1, \dots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

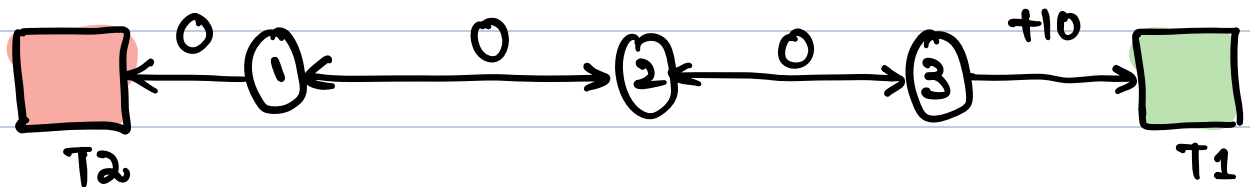
$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \epsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

Let's apply it on a random walk example.



States $\{1, 2, 3, T1, T2\}$

terminal states $\{T1, T2\}$.

Actions $\{L, R\}$.

Possible state actions $(1, L)$ $(1, R)$ $(2, L)$
 $(2, R)$ $(3, L)$ $(3, R)$.

$\Rightarrow$ Let $\epsilon = 0.01$.

Q	$(1, L)$	$(1, R)$	$(2, L)$	$(2, R)$	$(3, L)$	$(3, R)$
	0	0	0	0	0	0

R	(1,L)	
	(1,R)	
	(2,L)	
	(2,R)	
	(3,L)	
	(3,R)	

Generate an episode.

$S_2, L, 0, S_1, R, 0, S_2, R, 0, S_3, R, 10$
 we have $G=0$ $\gamma=+1$

Start from end.

(S_3, R)

$$G = G + \gamma R$$

$$G = 0 + 1(10)$$

$$\boxed{G = 10}$$

Do (S_3, R) exist on left? No.

So place this G in R matrix
 in row that corresponds to (S_3, R) .

move to Second (S,A) pair for left

$$(S_2, R). \quad G = 10 + \gamma(0) \\ G = 10.$$

Do (S_2, R) exist on left? No
So place G in (S_2, R) entry of
 R matrix.

Similarly for $(S_1, R) \quad G = 10$
 $(S_2, L) \quad G = 10.$

Now

<u>R</u>	1, L	
	1, R	10
	2, L	10
	2, R	10
	3, L	
	3, R	10

So

1, L	1, R	2, L	2, R	3, L	3, R

Q	0	10	10	10	0	10
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$$S_1 \quad A^* \rightarrow R$$

$$S_2 \quad A^* \rightarrow L, R \text{ (Let's select R).}$$

$$S_3 \quad A^* \rightarrow R$$

$$\text{for } S_1 \quad \pi(R|S_1) \quad 1 - 0.01 + \frac{0.01}{2}$$

$$\pi(L|S_1) \quad \frac{0.01}{2}$$

$$\text{Similarly for } S_2 \quad \pi(R|S_2) \quad 1 - 0.01 + \frac{0.01}{2}$$

$$\pi(L|S_2) \quad \frac{0.01}{2}$$

$$\text{Similarly } S_3 \quad \pi(R|S_3) \quad 1 - 0.01 + \frac{0.01}{2}$$

$$\pi(L|S_3) \quad \frac{0.01}{2}$$

Let's generate a new episode using this policy.

$S_2, L, 0, S_1, R, 0, S_2, L, 0, S_1, L, 0$

$$G_{1L} = 0$$

$$G_{1R} = 0$$

$$G_{2L} = 0$$

↑
we will not
use this return
in R matrix
bcz this pair
exist on
left.

Now R

1, L	0	
1, R	10	0
2, L	10	0
2, R	10	
3, L		
3, R.	10	

	1, L	1, R	2, L	2, R	3, L	3, R
Q	0	5	5	10	0	10

$S_1 \quad A^* \rightarrow R$
 $S_2 \quad A^* \rightarrow R$
 $S_3 \quad A^* \rightarrow R.$

again we will formulate the ϵ -soft
policy and continue till
policy does not change any
further

↓
Did you notice the size of R increasing
with every episode?

The reason is we have to keep
record of each value of G for
each state, action value so that
we can average it and get Q
value. The memory requirement thus
increases with each episode.

Let consider a case

the average of numbers is av.

Find average of 1, 2. $av = \frac{1+2}{2} = 1.5$

Find average if we also have 3.

$$av = \frac{1+2+3}{3} = 6/3 = 2.$$

Find average if we also have 4.

$$av = \frac{1+2+3+4}{4} = 10/4 = 2.5$$

Same issue: we have to keep all numbers in memory.

Solution

Let $av = 0$

first no is 1

$$\text{new. average} = \frac{\text{Pr. average} \times (n-1) + \text{number}}{n}$$

here $n = 1$ (no. of terms)

Pr. average = 0 (Previous average).

number = 1

$$\text{So new. average} = \frac{0 \times 0}{1} + \frac{1}{1} = 1.$$

Now we don't need to remember previous no.

we just remember average & no. of terms.

Now we have no 2

$$\begin{aligned}\text{new-average} &= \frac{1 \times (2-1)}{2} + \frac{2}{2} \\ &= \frac{1}{2} + 1 = 1.5\end{aligned}$$

Similarly for 3

$$\begin{aligned}\text{new-average} &= \frac{1.5 \times (3-1)}{3} + \frac{3}{3} \\ &= 0.5 \times 2 + 1 \\ &= 2.\end{aligned}$$

So we can just use formula

$$\text{new-average} = \frac{\text{Previous average} (n-1)}{\text{no. of terms}} + \frac{\text{new term}}{\text{no. of terms.}}$$

we can translate this to monte Carlo method.

Instead of storing each return in R matrix and then averaging, we can use old Q -value

$$Q_{\pi}(s, a) = \frac{Q_{\pi}(s, a) \times (n-1)}{n} + \frac{G_t}{n}$$

New Q -value

No. of first encounters

Return

This is called incremental Monte Carlo.

Now, we see that monte carlo is **not bootstrapping**. Instead, it is averaging return (in case of first visit) that are iid. Hence, its bias (systematic error) is zero. For every visit, although returns in a single episode are related, but itals. approaches zero bias as no. of episode increases.

So MC do not bootstrap.

Bias $\neq 0$.

On the other hand, as in MC we consider whole episode and episodes can vary in length and composition, hence MC is noisy and variance is high.

We know that

$$\text{Mean Square Error} = (\text{Bias})^2 + \text{Variance}$$

So MSE in MC is dominated by variance

To address high variance issue, we need to try other algorithm that do not want a complete episode to calculate value function.

We will discuss that algorithm i.e TD (Temporal Difference in next lecture).